

Homework 1

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Exercise 1

Let person 1 attach $2x$ units of value to her own income and x to person 2's income. The outcome $(1, 4)$ then has a payoff of $1 \cdot 2x + 4 \cdot x = 6x$. Similarly, $(2, 1)$ has a payoff of $2 \cdot 2x + 1 \cdot x = 5x$ and $(3, 0)$ has a payoff of $3 \cdot 2x + 0 \cdot x = 6x$.

$$(1, 4) = (3, 0) > (2, 1)$$

The payoff function $p(i_1, i_2) = 2i_1 + i_2$ would be consistent with these preferences.

Exercise 2

In the prisoner's dilemma, player 1 ranks the outcomes as following:

$$(C, N) > (N, N) > (C, C) > (N, C)$$

	X	Y
X	3, 3	1, 5
Y	5, 1	0, 0

In this game, player 1 would rank the outcomes as following:

$$(Y, X) > (X, X) > (X, Y) > (Y, Y)$$

This is not equivalent to the prisoner's dilemma, regardless of how the actions are named.

	X	Y
X	2, 1	0, 5
Y	3, -2	1, -1

In this game, player 1 would rank the outcomes as following:

$$(Y, X) > (X, X) > (Y, Y) > (X, Y)$$

This is equivalent to the prisoner's dilemma, with $Y = C$ and $X = N$.

Exercise 3

Let P be the action of choosing to insist on a preferred role and V be the action of mating in either role. We then have the following strategic game:

	V	P
V	$\frac{1}{2}(H + L), \frac{1}{2}(H + L)$	L, H
P	H, L	S, S

For this game to be equivalent to the prisoner's dilemma, it must be the case that for player 1:

$$(P, V) > (V, V) > (P, P) > (V, P)$$

with $C = P$ and $N = V$. Alternately, we could have had $C = V$ and $N = P$. For this to be true:

$$H > \frac{1}{2}(H + L) > S > L$$

Since $H > L$, this simplifies to:

$$\frac{1}{2}(H + L) > S > L$$

Exercise 4

Part (a)

Let Q be the quiet action and F be the fink action. In the case where $\alpha = 1$, we have the strategic game:

	Q	F
Q	4, 4	3, 3
F	3, 3	2, 2

In this game, player 1 would rank the outcomes as following:

$$(Q, Q) > (Q, F) = (F, Q) > (F, F)$$

This is not equivalent to the prisoner's dilemma, regardless of how the actions are named.

Part (b)

For an arbitrary $\alpha > 0$, we have the strategic game:

	Q	F
Q	$2 + 2\alpha, 2 + 2\alpha$	$3\alpha, 3$
F	$3, 3\alpha$	$1 + \alpha, 1 + \alpha$

For this game to be equivalent to the prisoner's dilemma, it must be the case that for player 1:

$$(F, Q) > (Q, Q) > (F, F) > (Q, F)$$

with $C = F$ and $N = Q$. Alternately, we could have had $C = Q$ and $N = F$. For this to be true:

$$3 > 2 + 2\alpha > 1 + \alpha > 3\alpha$$

Since $\alpha > 0$, this simplifies to:

$$3 > 2 + 2\alpha > 3\alpha$$

Consequently, $\alpha < 0.5$. When $\alpha = 0.5$, $(F, Q) = (Q, Q)$ and $(F, F) = (Q, F)$, so all four outcomes are Nash equilibria. When $\alpha > 0.5$, $(Q, Q) > (F, Q)$, so only (Q, Q) is a Nash equilibrium.